Reward Shaping and Terrain-Relative Termination for PPO Humanoid Locomotion Beyond Flat Ground

João Viterbo Vieira
Faculty of Engineering
University of Porto
Porto, Portugal
up202107689@edu.fe.up.pt

Francisco Pinto
Faculty of Engineering
University of Porto
Porto, Portugal
up199504009@edu.fe.up.pt

José Pedro Carvalho
SYSTEC-ARISE,
Faculty of Engineering
University of Porto
Porto, Portugal
jpcarvalho@fe.up.pt

António Pedro Aguiar
SYSTEC-ARISE,
Faculty of Engineering
University of Porto
Porto, Portugal
pedro.aguiar@fe.up.pt

***Abstract*—Humanoid locomotion is a challenging reinforcement learning problem due to high-dimensional actuation, intermittent contacts, and strong sensitivity to task and environment design. While policy-gradient methods such as Proximal Policy Optimization (PPO) reliably learn flat-ground walking, extending the same recipe to uneven terrain and long-horizon navigation frequently fails in practice, and the causes are often implementation-level rather than algorithmic. We present an empirical case study of two such design choices (reward alignment and episode-termination logic) using the MuJoCo Humanoid across four tasks of increasing difficulty: flat locomotion, stair climbing, waypoint-based circuit navigation, and circuits with stair segments. Task-aligned shaping (height progress and step milestones) yields a stair-climbing policy that reaches the top step in 96% of evaluation episodes with the smallest training budget of all four tasks. A controlled ablation shows fixed world-frame termination bounds truncating episodes of healthy agents on elevated terrain; consistent with this mechanism, on the combined circuit-with-stairs task none of nine development runs trained under such bounds learned stair traversal, whereas a terrain-relative health check enabled 87% circuit completion. All results come from single-training-seed runs with 100-episode evaluations and a five-seed evaluation-robustness sweep; environments, training configurations, and evaluation scripts are publicly released.**

***Index Terms*—Reinforcement Learning, Humanoid Locomotion, MuJoCo, Stair Climbing, Navigation.**

I. Introduction

Reinforcement learning (RL) provides a general framework for sequential decision-making in which an agent learns a policy that maximizes cumulative reward through interaction with an environment [1]. While RL has demonstrated strong performance on simulated continuous-control benchmarks, humanoid locomotion remains particularly challenging: policies that perform well on flat terrain frequently fail when extended to uneven terrain or long-horizon navigation.

In this work we focus on two implementation-level choices that repeatedly emerged as critical failure points when moving beyond flat locomotion: *reward design* and *termination criteria*. Reward functions that suffice for flat walking often provide weak or misleading learning signals on contact-rich terrain such as stairs. Similarly, termination logic tuned to a fixed ground reference can terminate episodes prematurely when the agent stands on elevated terrain, silently preventing learning.

This paper presents an empirical study of how reward design and episode-termination criteria affect humanoid locomotion learning beyond flat terrain, using the MuJoCo Humanoid [2], [3] and PPO [4] as implemented in Stable-Baselines3 [5]. We evaluate a sequence of tasks of increasing complexity: flat locomotion, stair climbing, waypoint-based circuit navigation, and circuits that include stair segments (Fig. 2). Rather than proposing a new algorithm, our objective is to identify which *implementation-level design decisions* most strongly affect learning stability and task success in this setting. Our contributions are empirical and practitioner-oriented:

- Two custom, fully parameterized MuJoCo environments: a configurable stairs environment and a waypoint-based circuit environment with optional stair segments.
- A structured comparison of four PPO configurations (S1–S4) that progressively introduce reward shaping, navigation structure, and terrain-aware termination handling.
- An evaluation protocol reporting task-completion success rates over 100 episodes per task.

For reproducibility, the full implementation (environments, training and evaluation scripts, and the exact per-run configurations) is publicly available at https://github.com/joao-viterbo-vieira/RL-Humanoid.

II. Related Work

Deep RL has produced locomotion behaviors for simulated humanoids from simple reward signals in rich environments [6], and reference-motion methods such as DeepMimic [7] demonstrated that seemingly minor implementation choices, notably *early termination*, can dominate learning outcomes for physics-based characters. Our study extends this line of implementation-focused analysis to terrain-aware termination in torque-controlled humanoid locomotion.

For legged robots, RL controllers trained in simulation have achieved agile and robust locomotion on challenging real terrain, typically using privileged terrain information or teacher–student distillation on quadrupeds [8]–[10] and massively parallel simulation [11], [12]. For bipeds, sim-to-real RL has produced blind stair traversal on Cassie [13], and causal-transformer policies trained with large-scale RL have controlled a full-scale humanoid outdoors [14]. These systems target hardware deployment with substantial engineering pipelines; in contrast, we deliberately isolate two design factors in a controlled, fully observable simulation setting on a standard benchmark humanoid, where their individual effect on learning can be observed directly.

Reward shaping has a long theoretical history [15], but task-specific shaping for contact-rich locomotion remains largely empirical; we document which shaping terms were necessary at each task level. Finally, our reporting follows the concerns raised about statistical rigor in deep RL evaluation [16], [17]: we state explicitly that final results come from one training seed per task and quantify evaluation variability with 100-episode success rates over multiple held-out evaluation seeds rather than small-sample reward statistics.

III. Problem Definition

We formalize each task as a Markov Decision Process (MDP) $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$: at each control step t the agent observes a state $s_t \in \mathcal{S}$, draws an action $a_t \in \mathcal{A}$ from a stochastic policy $\pi_\theta(a \mid s)$ with parameters θ, receives the scalar reward $r(s_t, a_t)$, and transitions to $s_{t+1} \sim P(\cdot \mid s_t, a_t)$, where P is determined by the MuJoCo physics configuration of each environment (Section IV).

Learning seeks the parameters θ that maximize the expected discounted return $\mathbb{E}_{\pi_\theta}[\sum_{t=0}^{\infty} \gamma^t\, r(s_t, a_t)]$, with discount $\gamma = 0.99$. The state-value function $V^\pi(s)$ is the expected return obtained by following π from state s, and the action-value function $Q^\pi(s, a)$ the expected return of taking action a in s and following π thereafter [1]. The PPO update (Section V-A) is driven by a sampled estimate $\hat{A}_t$ of the advantage $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$, which quantifies how much taking action a in state s improves on the policy's average behavior; V^π is approximated by a critic network acting as a variance-reducing baseline.

1) Simplified Robot Model and Task Variables: The humanoid is a torque-actuated articulated body, but apart from generic effort and contact-force penalties, all task-level quantities (reward terms and health checks) are defined on a reduced model of the torso alone. In a fixed world frame, the torso carries a position $\mathbf{p}_t = [x_t,\, y_t,\, z_t]^\top$, with height z_t, and an orientation given by the roll, pitch, and yaw angles $(\phi_t, \theta_t, \psi_t)$ extracted from the torso quaternion; the yaw ψ_t defines the heading (the time-indexed θ_t denotes pitch, distinct from the policy parameters θ). The planar linear velocity $\mathbf{v}_t = [\dot{x}_t,\, \dot{y}_t]^\top$ is the linear velocity of the torso root joint, read directly from the simulator state as the time derivative of $\mathbf{p}_t$; the flat-locomotion baseline instead follows the standard Gymnasium convention of finite-differencing the whole-body center of mass over the control interval. Relative to the heading, $\mathbf{v}_t$ decomposes into a component along the direction the agent faces and a component normal to it:

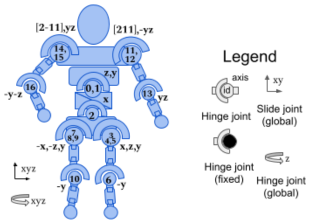


Fig. 1. Action space: the 17 torque-controlled joints of the MuJoCo Humanoid.

$$v_t^{\|} = \mathbf{v}_t^\top [\cos\psi_t,\, \sin\psi_t]^\top, \quad v_t^{\perp} = \mathbf{v}_t^\top [-\sin\psi_t,\, \cos\psi_t]^\top. \tag{1}$$

The stair course and circuit corridors are laid out along the world x-axis with centerline $y{=}0$, so when the agent faces down-course ($\psi_t \approx 0$) the world-frame forward velocity $\dot{x}_t$ coincides with $v_t^{\|}$ and the lateral offset y_t accumulates $v_t^{\perp}$ as a cross-track error. The reward terms of Section V-C are stated in these world-frame quantities to match the implementation; the distinction matters on course legs that leave the x-axis, which we flag there.

2) State Space: At each control step t the agent receives an observation o_t used as the state. The base observation is *proprioceptive*: joint positions and velocities, torso pose and velocities, and contact forces, following the standard MuJoCo Humanoid convention [3]. The global position (x_t, y_t) is excluded in all configurations, although the environment still uses it internally to evaluate rewards and health checks on the reduced model above. Tasks requiring exteroception augment the observation with a local terrain height grid and waypoint-relative navigation features (Section V-B).

3) Action Space: The action space is continuous, $a_t \in \mathbb{R}^{17}$: motor commands bounded by $[-0.4, 0.4]$ (dimensionless; mapped to joint torques via per-actuator gear ratios of 25–300) for the 17 actuated hinge joints of the humanoid (3 abdomen, 3 per hip, 1 per knee, 2 per shoulder, 1 per elbow; Fig. 1).

All environments use a physics timestep of $0.003\,\mathrm{s}$ with frame skip 5, i.e., a control interval of $dt = 0.015\,\mathrm{s}$ ($\approx 66.7\,\mathrm{Hz}$).

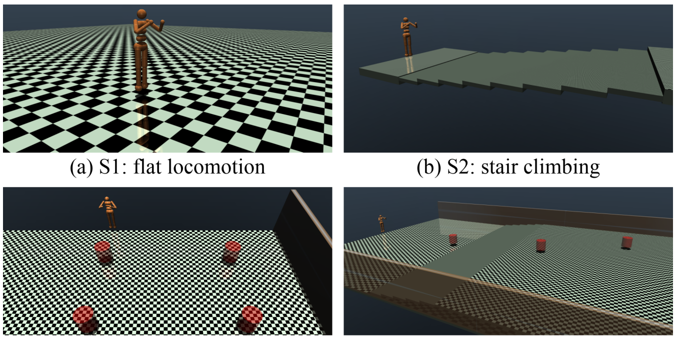
4) Episode Termination: Episodes end by *truncation* (time limit: 1000 steps for flat locomotion and stairs, 5000 for circuit tasks) or *termination* (health violation). Health is a function of torso height z_t: in the baseline, the agent is unhealthy when z_t leaves a fixed world-frame range $[z_{\min}, z_{\max}]$. On elevated terrain this produces false terminations even when the agent is standing upright. Configurations S2 and S4 therefore apply the health range to a *terrain-relative* height

$$z_{\mathrm{rel},t} = z_t - h(x_t, y_t), \tag{2}$$

where $h(x_t, y_t)$ is the terrain height at the agent's planar position.

IV. Environments and Tasks

We study four tasks of increasing difficulty (Fig. 2):



(a) S1: flat locomotion
(b) S2: stair climbing
(c) S3: circuit navigation (flat)
(d) S4: circuit with stairs

Fig. 2. Initial state of the four tasks, rendered with the exact geometry of the reported runs; translucent red cylinders mark the circuit waypoints.

1) **S1, Flat locomotion:** forward walking on flat ground (`Humanoid-v5`), the standard Gymnasium benchmark plus the shared shaping wrapper of Eq. (6), used as a control condition.
2) **S2, Stair climbing:** ascend a staircase (`HumanoidStairsConfigurable-v0`), parameterized by approach distance, number of steps, step height/depth, and top platform. The reported run uses 8 steps of height 0.10 m and depth 0.8 m; development runs spanned 8–15 steps and heights 0.10–0.20 m. An episode is successful if the agent reaches the top step.
3) **S3, Circuit navigation (flat):** visit a sequence of planar waypoints in order (`HumanoidCircuit-v0`). The reported course has four waypoints forming a rectangular loop with three turns; success requires reaching all waypoints.
4) **S4, Circuit with stairs:** the same environment with a three-waypoint course that includes a five-step stair segment (0.10 m rise) between the first two waypoints.

Side walls constrain circuit courses to a corridor, preventing trivial obstacle bypassing. Beyond flat locomotion, these tasks compound several difficulties: multi-objective coordination (progress, heading, obstacle traversal, energy), sparse milestone rewards atop dense locomotion signals, contact discontinuities at stair edges, and accurate turning.

All custom environments use MuJoCo's `RK4` integrator with the `PGS` solver (50 iterations), terrain friction $(1.0, 0.1, 0.1)$, and `condim=3` terrain contacts; unspecified options remain at MuJoCo defaults.

V. Method

A. Algorithm

We use PPO [4] from Stable-Baselines3 [5], an on-policy actor–critic method with the clipped surrogate objective

$$L^{\mathrm{CLIP}}(\theta) = \hat{\mathbb{E}}_t\Big[\min\Big(\rho_t(\theta)\,\hat{A}_t,\ \mathrm{clip}(\rho_t(\theta),\,1{-}\epsilon,\,1{+}\epsilon)\,\hat{A}_t\Big)\Big], \tag{3}$$

where $\rho_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\mathrm{old}}}(a_t \mid s_t)$ is the importance-sampling ratio and $\hat{A}_t$ is the GAE estimate of the advantage (Section III; $\lambda = 0.95$, $\epsilon = 0.2$). Table I lists the training hyperparameters as actually used in the reported runs.

TABLE I
PPO TRAINING HYPERPARAMETERS OF THE REPORTED RUNS.

Parameter	**S1**	**S2–S4**
Network (MLP, Tanh)	256×256	256×256
Learning rate	2.5×10^{-4}	3×10^{-4}
Rollout length	1,024 $\times$ 16 envs	4,096 $\times$ 8 envs
Minibatch size	8,192	16,384
Epochs per update	10	10
Entropy / VF coef.	0.005 / 0.5	0.005 / 0.25
Grad. clip / γ / GAE λ	0.5 / 0.99 / 0.95	0.5 / 0.99 / 0.95
Normalization	VecNormalize (obs. & rewards)	
Training budget	50M steps	30M–80M steps

TABLE II
OBSERVATION SPACE DIMENSIONS PER CONFIGURATION.

Component	**S1**	**S2**	**S3**	**S4**
Proprioceptive core	348	348	348	348
Additional proprioceptive signals	—	28	28	28
Terrain grid g_t	—	25	25	25
Waypoint features w_t	—	—	5	5
Total	**348**	**401**	**406**	**406**

The 28 additional channels are world-body inertia, COM velocity, contact-force, and root actuation signals.

B. Observation Design

All configurations share the proprioceptive base observation. Task-specific augmentations are introduced progressively.

1) Terrain Grid (S2–S4): A local terrain map of 25 samples (5×5 offsets) centered on the agent:

$$g_t(i, j) = h(x_t + \delta_i,\, y_t + \delta_j) - z_t, \tag{4}$$

with fixed world-frame offsets $\delta_i, \delta_j \in \{-0.6, -0.3, 0, 0.3, 0.6\}$ m; heights are encoded relative to the agent's current z_t.

2) Waypoint Features (S3, S4): Navigation state is a 5-dimensional vector

$$w_t = \Big[\,\Delta x_t,\ \Delta y_t,\ \tfrac{n_t}{N_{\mathrm{wp}}},\ \sin(\epsilon_t),\ \cos(\epsilon_t)\Big]^\top, \tag{5}$$

where $[\Delta x_t,\ \Delta y_t]^\top$ is the displacement to the active waypoint, n_t/N_{wp} the fraction of completed waypoints, and $\epsilon_t = \mathrm{atan2}(\Delta y_t, \Delta x_t) - \psi_t$, wrapped to $(-\pi, \pi]$, the heading error between the yaw ψ_t (Section III) and the waypoint direction; $\sin / \cos$ encoding avoids the angle-wrap discontinuity at $\pm\pi$. Table II summarizes observation dimensionality.

C. Reward Design

The reward optimized by PPO is a task-specific environment reward r_t^{env} plus a shaping wrapper common to all configurations:

$$r_t^{\mathrm{total}} = r_t^{\mathrm{env}} - \underbrace{w_e \|a_t\|_2^2}_{\text{effort}} - \underbrace{w_\ell |y_t|}_{\text{lateral}}, \tag{6}$$

where y_t is the lateral offset of Section III-1, penalizing world-frame drift from the corridor centerline $y{=}0$. Table III collects every weight and task-specific value of this section; the prose below only defines the terms.

1) S1: Baseline: The standard `Humanoid-v5` reward,

$$r_t^{\text{env}} = \underbrace{w_h \mathbb{I}[z_{\min} < z_t < z_{\max}]}_{\text{healthy}} + \underbrace{w_f \dot{x}_t}_{\text{forward}} - \underbrace{w_c \|a_t\|_2^2}_{\text{ctrl}} - c_t^{\text{cf}}, \quad (7)$$

where $\mathbb{I}[\cdot]$ is the indicator function (1 if its argument holds, 0 otherwise) and the contact-force cost $c_t^{\text{cf}} = w_{\text{cf}} \min(\|F_{\text{ext}}\|_2^2, 10)$ is saturated so rare impact spikes cannot dominate. The forward term pays the world-frame velocity $\dot{x}_t$ of Section III-1, i.e., the heading-aligned velocity $v_t^{\parallel}$ for an agent facing down-course.

2) S2: Stairs: Climbing-specific shaping is added:

$$r_t^{\text{env}} = r_t^{\text{forward}} + r_t^{\text{height}} + r_t^{\text{step}} + r_t^{\text{healthy}} - c_t^{\text{ctrl}} - c_t^{\text{cf}}, \quad (8)$$

where $r_t^{\text{height}} = w_z \max(0, z_t - z_{t-1})$ rewards upward progress and $r_t^{\text{step}} = b_s \Delta n_t$ is a milestone bonus for each newly reached stair step. Relative to S1, the reported run strengthens the forward and healthy terms, softens the contact-force cost, and applies the terrain-relative health check of Eq. (2).

3) S3: Waypoint Navigation (Flat): Forward velocity is replaced by navigation-aligned signals:

$$r_t^{\text{env}} = r_t^{\text{prog}} + r_t^{\text{dir}} + r_t^{\text{head}} + r_t^{\text{bal}} + r_t^{\text{speed}} + r_t^{\text{bonus}} + r_t^{\text{healthy}} - c_t^{\text{ctrl}} - c_t^{\text{cf}}, \quad (9)$$

with progress $r_t^{\text{prog}} = w_p(d_{t-1} - d_t)$, where d_t is the planar distance to the active waypoint; directional velocity $r_t^{\text{dir}} = w_f \max(0, \mathbf{v}_t \cdot \hat{\mathbf{u}}_t)$, projecting the planar velocity of Section III-1 onto the unit direction $\hat{\mathbf{u}}_t$ to the waypoint; heading alignment $r_t^{\text{head}} = w_\psi \cos(\epsilon_t) \min(\|\mathbf{v}_t\|, 2)$, which for velocity aligned with the heading equals the saturated velocity component toward the waypoint (both terms active only beyond 0.1 m of the waypoint); and sparse bonuses $r_t^{\text{bonus}} = b_{\text{wp}} \mathbb{I}[d_t < \tau] + b_{\text{circ}} \mathbb{I}[\text{all waypoints reached}]$ for entering the active waypoint's reach radius τ and for completing the course. Two regularizers stabilize multi-objective learning:

$$r_t^{\text{bal}} = w_b \cos\left(\sqrt{\phi_t^2 + \theta_t^2}\right), \quad r_t^{\text{speed}} = w_v e^{-\frac{1}{2}\left(\frac{\|\mathbf{v}_t\| - v^\star}{\sigma_v}\right)^2}, \quad (10)$$

rewarding small torso tilt via the roll and pitch of Section III-1 and planar speeds near a target $v^\star$. One imperfection is retained for cross-task comparability: the wrapper's lateral term penalizes the world-frame offset $|y_t|$ rather than the heading-relative cross-track error of Section III-1, so it opposes the navigation objective on course legs whose waypoints lie off-axis. We revisit its likely effect in Section VI.

4) S4: Circuit with Stairs: S4 uses the S3 reward with three changes for elevated terrain: the height-progress term from S2 is reintroduced at a smaller weight, the speed target $v^\star$ is lowered, and, critically, termination uses the terrain-relative check of Eq. (2).

TABLE III
REWARD TERMS AND PER-CONFIGURATION VALUES. THE HEALTHY RANGE PARAMETERIZES BOTH THE HEALTHY BONUS AND TERMINATION (SECTION III-4); [r] MARKS RANGES APPLIED TO THE TERRAIN-RELATIVE HEIGHT OF EQ. (2).

Term		**S1**	**S2**	**S3**	**S4**
Healthy bonus	w_h	5	15	5	5
Forward velocity	w_f	1.25	2	—	—
Height progress	w_z	—	8	—	2
Step milestone	b_s	—	5	—	—
Waypoint progress	w_p	—	—	200	200
Directional velocity	w_f	—	—	1	1
Heading alignment	w_ψ	—	—	2	2
Balance	w_b	—	—	0.5	0.5
Speed regulation	w_v	—	—	0.2	0.2
target / width (m/s)	$v^\star/\sigma_v$	—	—	1.2 / 0.5	1.0 / 0.5
Waypoint bonus	b_{wp}	—	—	150	150
reach radius (m)	τ	—	—	1.5	1.5
Circuit bonus	b_{circ}	—	—	500	500
Control cost	w_c	0.1	0.1	0.1	0.1
Contact force ($\times 10^{-7}$)	w_{cf}	5	2	5	5
Wrapper effort	w_e	0.001	0.001	0.001	0.001
Wrapper lateral	w_ℓ	1.0	0.3	1.0	1.0
Healthy range (m)		$[1, 2]$	$[1, 2]^{\text{r}}$	$[0.8, 3]$	$[1, 2]^{\text{r}}$

TABLE IV
FINAL TRAINING RUNS (SEED 42) AND EVALUATION OVER 100 EPISODES PER TASK (DETERMINISTIC POLICY, HELD-OUT EVALUATION SEED).

Task	**Steps**	**Ckpt.**	**Reward**	**Len.**	**Success**[a]
S1: Flat	50M	42M	$8{,}429 \pm 2{,}320$	912	84%
S2: Stairs	30M	29M	$15{,}760 \pm 1{,}868$	960	**96%**
S3: Circuit	80M	80M	$6{,}949 \pm 1{,}537$	514	47%
S4: Circ.+St.	80M	64M	$10{,}134 \pm 2{,}421$	598	**87%**

Ckpt.: timestep of the best evaluation checkpoint. Len.: mean episode length (control steps). Success: task-defined completion rate (see text).
[a] Robustness sweep over five additional held-out evaluation seeds (seeds 0–4, 100 episodes each): deterministic success 80 ± 5, 92 ± 3, 54 ± 5, 81 ± 5 % for S1–S4 (mean $\pm$ s.d.), in line with the single-seed values above.

VI. RESULTS

A. Protocol

Development comprised 58 training runs totaling approximately 2.6B environment steps, through which rewards, observations, and termination logic were iterated. The *final* results below come from one training run per task (Table IV), each trained independently with seed 42; incremental training across tasks did not reliably preserve previously learned behaviors (Section VII). For each task we evaluate the best checkpoint for 100 episodes with a held-out evaluation seed, observation normalization frozen to the training statistics, and rewards reported in raw (unnormalized) units. We report the deterministic policy (mode actions); stochastic-policy results are noted in the text. Success is task-defined: no fall over the full 1000-step horizon (S1), reaching the top step (S2), and reaching all waypoints (S3/S4). Circuit episodes do not terminate on completion: success is recorded once all waypoints are reached and the episode continues until a fall or truncation, which keeps post-completion stability observable (a fall after completion does not revoke success).

B. Flat Locomotion (S1)

The baseline solves flat walking: 84% of the 100 evaluation episodes complete the full 1000-step horizon without a fall (mean reward $8{,}429 \pm 2{,}320$). The training curve (Fig. 3, top

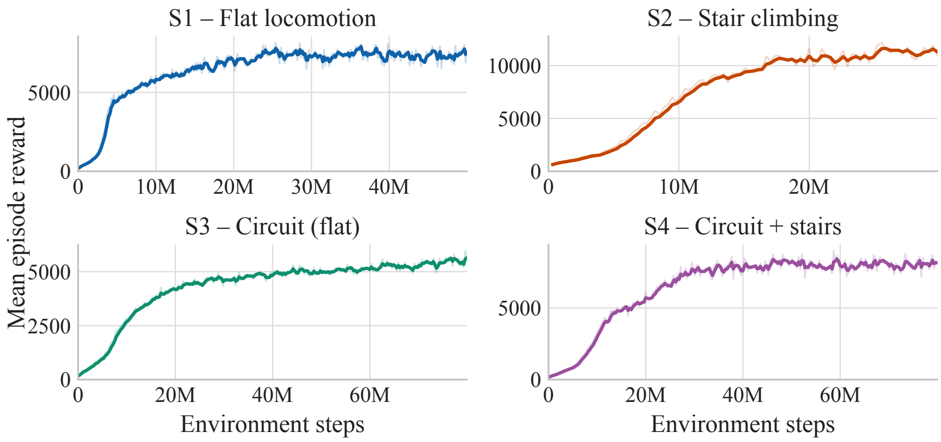


Fig. 3. Training curves for the four final runs: mean episode return per rollout batch in raw reward units (faded: raw; solid: EMA-smoothed). Reward scales are task-specific and not comparable across panels.

left) shows rapid initial learning in the first 10M steps and a plateau around 7,000–7,600 from roughly 25M steps; the best checkpoint was selected at 42M of 50M steps. Under stochastic actions the success rate drops to 30%, indicating that the learned gait is sensitive to action noise.

C. Stair Climbing (S2)

With climbing-aligned shaping, the agent reaches the top step in 96% of episodes (mean reward $15{,}760 \pm 1{,}868$, mean length 960) and then walks stably on the platform. Training (Fig. 3, top right) is the steepest of all configurations: near-asymptotic performance by $\sim$20M steps with the best checkpoint at 29M of a 30M budget, the smallest budget among all tasks despite the contact-rich terrain. We attribute this to the dense, well-aligned signal provided by height-progress and step-milestone terms on top of terrain-relative termination.

D. Circuit Navigation, Flat (S3)

The flat circuit is only partially mastered: the policy reaches on average 3.3 of 4 waypoints, but completes the full course in only 47% of episodes (mean reward $6{,}949 \pm 1{,}537$; mean length 514 steps). Failures concentrate in the later, sharper turns: precisely the course legs where the wrapper's centerline penalty (Section V-C) opposes progress, a misalignment we disclose as a likely contributing factor. Training (Fig. 3, bottom left) rises steadily across the entire 80M-step budget without a clear plateau, and the best checkpoint coincides with the final timestep, suggesting the policy had not converged; the sparse waypoint bonuses provide a weaker signal than the dense forward-velocity and height-progress rewards of S1/S2. Extending training was precluded by the compute budget (Section VII-A).

E. Circuit with Stairs (S4)

The combined task reaches 87% full-circuit completion, including the stair segment, at a mean reward of $10{,}134 \pm 2{,}421$ and 2.7 of 3 waypoints on average. Training (Fig. 3, bottom right) plateaus near 8,000 from $\sim$30M steps with persistent oscillations; the best checkpoint was reached at 64M of 80M steps. We attribute the oscillations to the multi-objective composite reward, in which small degradations of one component (e.g., stair traversal) temporarily depress the total. Note that S4's higher success rate relative to S3 does not imply stairs make navigation easier: the S4 course has three waypoints and one sharp turn versus four waypoints and three turns in S3, so the two success rates are not directly comparable.

Crucially, this configuration *did not learn at all* under fixed world-frame termination: across nine development runs ($\sim$520M steps), none traversed the stair segment, as episodes terminated the moment the agent stood on elevated terrain. The terrain-relative check of Eq. (2) was the single change that enabled learning; Section VI-F isolates this mechanism in a controlled evaluation-time ablation.

F. Termination Ablation

To isolate the termination mechanism under controlled conditions, we re-evaluated the trained S2 and S4 policies (both trained with the terrain-relative check) for 100 episodes under each termination rule, all else fixed (Table V). On S2 the fixed world-frame check leaves success unchanged (96%; the top step is reached before the check fires) yet cuts mean episode length from 960 to 645 steps and halts the agent at $x{=}7.9$ m rather than 12.7 m; every such termination occurs while the agent stands on the elevated platform (mean terrain height 0.70 m), i.e., spurious rather than a genuine fall. During training, removing exactly these on-platform transitions is what starves the policy of the experience needed to learn stair traversal. The effect is muted on S4 (88% vs. 87%), whose course reaches lower elevations (mean 0.46 m) that trip the fixed bound less often. The ablation confirms the mechanism on a competent policy; the stronger claim that world-frame bounds *prevent learning* rests on the development-run evidence above.

G. Convergence and Robustness

Two cross-task patterns stand out. First, convergence speed tracks reward alignment rather than task simplicity: S2 (stairs)

TABLE V

EVALUATION-TIME TERMINATION ABLATION: TRAINED S2/S4 POLICIES RE-EVALUATED FOR 100 EPISODES UNDER EACH RULE. WORLD-FRAME BOUNDS FIRE ON ELEVATED TERRAIN, TRUNCATING S2 WELL SHORT OF A FALL.

Task	Termination	Success	Len.	Final x
S2: Stairs	terrain-relative	96%	960	12.7
S2: Stairs	world-frame	96%	**645**	**7.9**
S4: Circ.+St.	terrain-relative	87%	598	18.8
S4: Circ.+St.	world-frame	88%	618	18.9

converged fastest despite terrain interaction, while S3 (flat circuit) was slowest, consistent with its sparser milestone signal. Second, training-curve smoothness is a poor predictor of task success: S3 has the smoothest curve but the lowest completion rate (47%), whereas S4's oscillatory curve yields 87%. Deterministic-vs-stochastic gaps (84% vs. 30% on S1; 96% vs. 52% on S2; 47% vs. 15% on S3; 87% vs. 37% on S4) further caution against judging policies by a single evaluation mode.

VII. CONCLUSIONS

We studied how implementation-level design choices govern the transition of PPO-trained humanoid agents from flat locomotion to stair climbing and long-horizon waypoint navigation. Our results show that no algorithmic novelty was needed to move beyond flat ground. Once two design factors, reward alignment and termination logic, were brought in line with the task, a standard PPO recipe reached the top of the staircase in 96% of evaluation episodes and completed the combined circuit-with-stairs course in 87%.

The first factor is reward alignment. Forward-velocity rewards suffice on flat ground but do not describe progress on stairs; explicit height-progress and step-milestone terms made stair climbing the most reliably solved task of the four within the smallest training budget. Conversely, the sparse waypoint bonuses of the circuit task yielded the slowest learning and the lowest completion rate, and the retained world-frame centerline penalty (Section V-C) illustrates how a single misaligned shaping term can quietly work against the task objective.

The second factor is termination logic: the controlled ablation of Section VI-F shows fixed world-frame health bounds spuriously terminating healthy agents on elevated terrain, and none of nine development runs trained under those bounds learned stair traversal; the terrain-relative check of Eq. (2), a small and generally applicable change, was the single modification that unlocked the combined task, echoing early-termination observations in character animation [7]. A third, cautionary observation concerns curricula: continuing training from a stair-climbing policy into circuit navigation did not preserve earlier skills, a repeated qualitative pattern from development runs rather than a controlled comparison, consistent with known limitations of naive curricula [18]. Final results therefore use independent task-specific runs.

A. Limitations and Future Work

All final results come from a single training seed per task; given known seed sensitivity in deep RL [16], [17], the quantitative values should be read as a documented case study rather than population estimates (the five-seed sweep in Table IV probes evaluation noise only, not training-seed variance), and the termination finding as a combination of repeated development-run observations and the controlled evaluation-time ablation of Table V. Each run required roughly 3 hours, and iterating rewards and hyperparameters across four tasks consumed the available budget (Section VI), precluding per-task sweeps.

These limitations chart a clear path forward: the immediate next step is a multi-seed training campaign, including a training-time termination ablation to complement the evaluation-time one reported here, followed by replacing the privileged height grids with depth-based perception, domain randomization over terrain geometry, and skill-preservation mechanisms (modular or constrained policies) for multi-task learning. We open-source all environments, configurations, and evaluation scripts to support these further developments.

REFERENCES

[1] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA: MIT Press, 2018.

[2] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.

[3] M. Towers *et al.*, "Gymnasium: A standard interface for reinforcement learning environments," arXiv preprint arXiv:2407.17032, 2024.

[4] J. Schulman *et al.*, "Proximal policy optimization algorithms," arXiv preprint arXiv:1707.06347, 2017.

[5] A. Raffin *et al.*, "Stable-Baselines3: Reliable reinforcement learning implementations," *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.

[6] N. Heess *et al.*, "Emergence of locomotion behaviours in rich environments," arXiv preprint arXiv:1707.02286, 2017.

[7] X. B. Peng *et al.*, "DeepMimic: Example-guided deep reinforcement learning of physics-based character skills," *ACM Transactions on Graphics*, vol. 37, no. 4, pp. 143:1–143:14, 2018.

[8] J. Hwangbo *et al.*, "Learning agile and dynamic motor skills for legged robots," *Science Robotics*, vol. 4, no. 26, p. eaau5872, 2019.

[9] J. Lee *et al.*, "Learning quadrupedal locomotion over challenging terrain," *Science Robotics*, vol. 5, no. 47, p. eabc5986, 2020.

[10] T. Miki *et al.*, "Learning robust perceptive locomotion for quadrupedal robots in the wild," *Science Robotics*, vol. 7, no. 62, p. eabk2822, 2022.

[11] N. Rudin *et al.*, "Learning to walk in minutes using massively parallel deep reinforcement learning," in *Proc. 5th Conf. Robot Learning (CoRL)*, 2022, pp. 91–100.

[12] V. Makoviychuk *et al.*, "Isaac Gym: High performance GPU-based physics simulation for robot learning," arXiv preprint arXiv:2108.10470, 2021.

[13] J. Siekmann *et al.*, "Blind bipedal stair traversal via sim-to-real reinforcement learning," in *Robotics: Science and Systems XVII (RSS)*, 2021.

[14] I. Radosavovic *et al.*, "Real-world humanoid locomotion with reinforcement learning," *Science Robotics*, vol. 9, no. 89, p. eadi9579, 2024.

[15] A. Y. Ng, D. Harada, and S. Russell, "Policy invariance under reward transformations: Theory and application to reward shaping," in *Proc. 16th Int. Conf. Machine Learning (ICML)*, 1999, pp. 278–287.

[16] P. Henderson *et al.*, "Deep reinforcement learning that matters," in *Proc. 32nd AAAI Conf. Artificial Intelligence*, 2018, pp. 3207–3214.

[17] R. Agarwal *et al.*, "Deep reinforcement learning at the edge of the statistical precipice," in *Adv. Neural Information Processing Systems (NeurIPS)*, 2021.

[18] Y. Bengio *et al.*, "Curriculum learning," in *Proc. 26th Int. Conf. Machine Learning (ICML)*, 2009, pp. 41–48.